

Concepts

① Linear model vs. non-linear.

$y \rightarrow$ response $x_1 \dots x_p \rightarrow$ predictors

$$y = f(x_1 \dots x_p) + \epsilon \quad \text{noise is additive}$$

f is linear in the parameters.

f may involve multiple unknown parameters.

② Causal relation: Having a good linear model fit does not imply a causal relationship between the variables.

"All models are wrong, but some are useful." — Box

③ Interpretation of the coefficients.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon.$$

β_0 : ^{expected} value of y when all x 's are zero.

β_1 : expected change in y when x_1 changes by 1 unit keeping all other ~~x~~ predictors fixed.

Think about Simpson's paradox when several predictors are allowed to change together.

④ Collinearity: A linear relationship between the predictors.

One column of the $X_{n \times (p+1)}$ is a linear combination of some other columns: x 's exhibit collinearity.

eg $p=2$. No column of 1's (intercept).

$$x_1 = 2x_2 \quad (\text{exact linear relation})$$

$$y = x_1 + x_2 + \epsilon \quad \text{is the same model as} \quad y = 3x_2 + \epsilon.$$

Parameter $\beta = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$ or $\begin{pmatrix} 1.5 \\ 0 \end{pmatrix}$ are all correct
non-identifiability.

Estimation $\hat{\beta} = (X^T X)^{-1} X^T y$ $X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{bmatrix}$ ~~$x_1 = 2x_2$~~ $x_1 = 2x_2$

$\text{rank}(X) = 1$ $\text{rank}(X^T X) = 1$.

$$\text{rank}(X) = 1 \quad \text{rank}(X^T X) = 1.$$

$X^T X$ does not have an inverse.

No least squares estimate.

Solution: Throw away one of the predictors.

With $p > 2$ how do you find if there is collinearity.

$$x_1 = 3x_2 + 1.7x_3 + 3.5x_4 + \eta \quad x_5$$

Check if $X^T X$ is singular by calculating the determinant.

Multiple collinearities are possible.

$$A = (X^T X) \quad A^{-1} = \frac{1}{\det A} \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \quad \text{cofactor matrix.}$$

If diagonal elements are zero, then keep corresponding columns.

In general predictors are observed with noise.

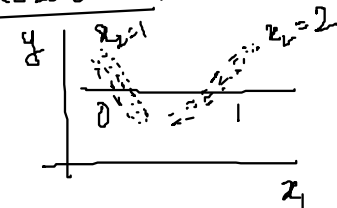
$X^T X$ is not exactly singular but nearly so. (near collinearity)

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}.$$

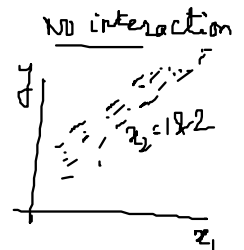
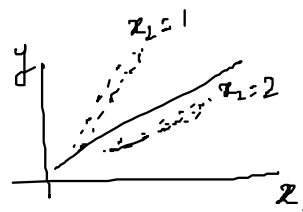
When $\det(X^T X) \approx 0$, then $(X^T X)^{-1}$ has very large elements.

There is a large variation in parameter estimates.

⑤ Interaction



~~graph~~



eg. $y \rightarrow$ salary
 $x_1 \rightarrow$ experience
 $x_2 \rightarrow$ gender.

Introduce more terms



$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon.$$

$I(x_2=2)$

⑥ Normal equation:

$$y = X\beta + \epsilon$$

perpendicular

$X\hat{\beta}$ is the orthogonal projection of

Y in the space of X .

Least squares: minimizing $(Y - X\beta)^T(Y - X\beta) = E^T E$.

Chapter 3

Inference for Linear Regression

Model $Y = X\beta + E$ $n \times 1$ $n \times (p+1)$ $(p+1) \times 1$ $n \times 1$ p predictors + intercept.

(New) Assume: E_i are indep and identically distributed as $N(0, \sigma^2)$

X fixed & known

Y is observed data.

β, σ^2 parameters

$$E \sim N_n(0, \sigma^2 I)$$

Normality is the new assumption.

Goal: MLE of β, σ^2 . [$p+2$ parameters]

$$L(\beta, \sigma^2 | y) = \frac{1}{(2\pi)^{n/2} \sigma^n} \exp\left\{-\frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta)\right\}$$

$$f(E) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left\{-\frac{1}{2} \left(\frac{E_i - 0}{\sigma}\right)^2\right\}$$

$$= \frac{1}{(2\pi)^{n/2} \sigma^n} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n E_i^2\right\}$$

$$= \frac{1}{(2\pi)^{n/2} \sigma^n} \exp\left\{-\frac{1}{2\sigma^2} E^T E\right\}$$

$$\ell(\beta, \sigma^2 | y) = -\frac{n}{2} \ln 2\pi - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta)$$

Max ℓ w.r.t β with fixed σ is same as $\min (Y - X\beta)^T (Y - X\beta)$

This is same as the least squares criterion.

So we get the least squares estimates by taking

$$\frac{\partial \ell}{\partial \beta} = 0. \quad \text{w.} \quad \hat{\beta}_{MLE} = \hat{\beta}_{LS} = (X^T X)^{-1} X^T Y.$$

$$\frac{\partial \ell}{\partial \sigma^2} = 0. \quad \frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2} \cdot \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} (Y - X\hat{\beta})^T (Y - X\hat{\beta})$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} (Y - X\hat{\beta}_{MLE})^T (Y - X\hat{\beta}_{MLE})$$

$$= \frac{1}{n} E^T E$$

$\hat{\sigma}_{MLE}^2$ is a biased estimate of σ^2 .

$$\text{Claim: } E[\hat{\sigma}_{MLE}^2] = \frac{n-p-2}{n} \sigma^2$$

$\hat{\sigma}_{MLE}^2$ is a biased estimate of σ^2

Claim $E(\hat{\sigma}_{MLE}^2) = \frac{n-(p+1)}{n} \sigma^2$

$$\hat{\sigma}_{MLE}^2 \cdot \frac{n}{n-(p+1)} = s^2 = \frac{1}{n-(p+1)} \hat{E}^T \hat{E}$$

↓
is unbiased for σ^2 .

Pf of claim : $\hat{E} = Y - X\hat{\beta}$

$$= Y - X(X^T X)^{-1} X^T Y \quad [Y = X\beta + \varepsilon, \hat{\beta} = (X^T X)^{-1} X^T Y]$$

$$= X\beta + \varepsilon - X(X^T X)^{-1} X^T X\beta - X(X^T X)^{-1} X^T \varepsilon$$

$$= X\beta + \varepsilon - X\beta - X(X^T X)^{-1} X^T \varepsilon$$

$$= \varepsilon - X(X^T X)^{-1} X^T \varepsilon$$

$$= [I - X(X^T X)^{-1} X^T] \varepsilon = A \varepsilon$$

$$\hat{E} = A \varepsilon \quad \text{with } A = I - X(X^T X)^{-1} X^T$$

$$\varepsilon \sim N_n(0, \sigma^2 I)$$

$$\hat{E} \sim N(A \cdot 0, A \sigma^2 I A^T)$$

" " $\sigma^2 A$

$$\left[\frac{1}{\sigma} \hat{E} \right] \sim N_n(0, [I - X(X^T X)^{-1} X^T])$$

$$\left(\frac{1}{\sigma} \hat{E} \right)^T \left(\frac{1}{\sigma} \hat{E} \right) \sim \chi_k^2$$

$$\frac{1}{\sigma^2} \hat{E}^T \hat{E} \sim \chi_k^2 \quad \text{where } k = \text{rank}(I - X(X^T X)^{-1} X^T)$$

(*) For idempotent matrix A , $\text{rank}(A) = \text{tr}(A)$

$$k = \text{rank}(I - X(X^T X)^{-1} X^T)$$

$$= \text{tr}(I - X(X^T X)^{-1} X^T) = \text{tr}(I) - \text{tr}(X(X^T X)^{-1} X^T)$$

$$= n - \text{tr}[(X^T X)^{-1} X^T X] = n - \text{tr}(I_{p+1}) = n - (p+1)$$

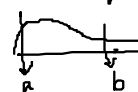
$$\frac{1}{\sigma^2} \hat{E}^T \hat{E} \sim \chi_{n-(p+1)}^2$$

$$s^2 = \frac{1}{n-(p+1)} \hat{E}^T \hat{E}$$

$$= \frac{\sigma^2}{n-(p+1)} \left[\frac{1}{\sigma^2} \hat{E}^T \hat{E} \right]$$

$$\left[s^2 \sim \frac{\sigma^2}{n-(p+1)} \chi_{n-(p+1)}^2 \right]$$

For CI for σ^2
(a, b) → 5% & 95% critical values / quantiles of $\chi_{n-(p+1)}^2$



$$\boxed{s^2 \sim \frac{\sigma^2}{n-(p+1)} \chi^2_{n-(p+1)}} \quad \begin{matrix} a & b \end{matrix}$$

$$E(s^2) = \sigma^2 \quad [E(\chi^2_k) = k]$$

Distribution of $\hat{\beta}$

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T Y \\ &= (X^T X)^{-1} X^T X \beta + (X^T X)^{-1} X^T E \\ &= \beta + (X^T X)^{-1} X^T E. \end{aligned} \quad E \sim N(0, \sigma^2 I)$$

Prob result $\boxed{\text{If } E \sim N(\mu, \Sigma) \quad AE \sim N(A\mu, A\Sigma A^T)}$
 then $AE + b \sim N(A\mu + b, A\Sigma A^T)$

$$\hat{\beta} \sim N_{p+1}(\beta, \sigma^2 (X^T X)^{-1})$$

CI for σ^2 $s^2 \sim \frac{\sigma^2}{n-(p+1)} \chi^2_{n-(p+1)}$

s^2 is a statistic $s^2 = \frac{1}{n-(p+1)} \sum \hat{\epsilon}_i^2$, $\hat{\epsilon}_i = y_i - \hat{y}_i$
 $\hat{\epsilon}_i = y_i - \hat{y}_i$

(a, b) are 2.5% & 97.5% quantiles of $\chi^2_{n-(p+1)}$ distn.

$$P[a < \chi^2_{n-(p+1)} < b] = .95$$

$$P\left[a < \frac{s^2 (n-(p+1))}{\sigma^2} < b\right] = .95$$

$$P\left[\frac{s^2 (n-(p+1))}{b} < \sigma^2 < \frac{s^2 (n-(p+1))}{a}\right] = .95$$

CI for β_i $\hat{\beta} \sim N_{p+1}(\beta, \sigma^2 (X^T X)^{-1})$

suppose we know σ^2 , $\hat{\beta}_i \sim N(\beta_i, \sigma^2 [(X^T X)^{-1}]_{ii})$
 unrealistic

$$\frac{\hat{\beta}_i - \beta_i}{\sigma \sqrt{[(X^T X)^{-1}]_{ii}}} \sim N(0, 1)$$

$$P\left(\hat{\beta}_i - z_{\alpha/2} \sigma \sqrt{[(X^T X)^{-1}]_{ii}} < \beta_i < \hat{\beta}_i + z_{\alpha/2} \sigma \sqrt{[(X^T X)^{-1}]_{ii}}\right) = 1 - \alpha.$$

σ unknown

Replace with $\hat{\sigma}$.

But $\frac{\hat{\beta}_i - \beta_i}{\hat{\sigma} \sqrt{[(X^T X)^{-1}]_{ii}}} \not\sim N(0, 1).$

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N\left(\frac{\bar{X} - \mu}{s/\sqrt{n}}, 1\right)$$

$$\boxed{\text{indep}} \leftarrow \frac{N}{\sqrt{\chi^2/d_f}} \sim t$$

$\hat{\sigma}^2 = s^2$ is indeed a multiple
of χ^2 . (const)

To show $\hat{\beta}$ and s^2 are indep.